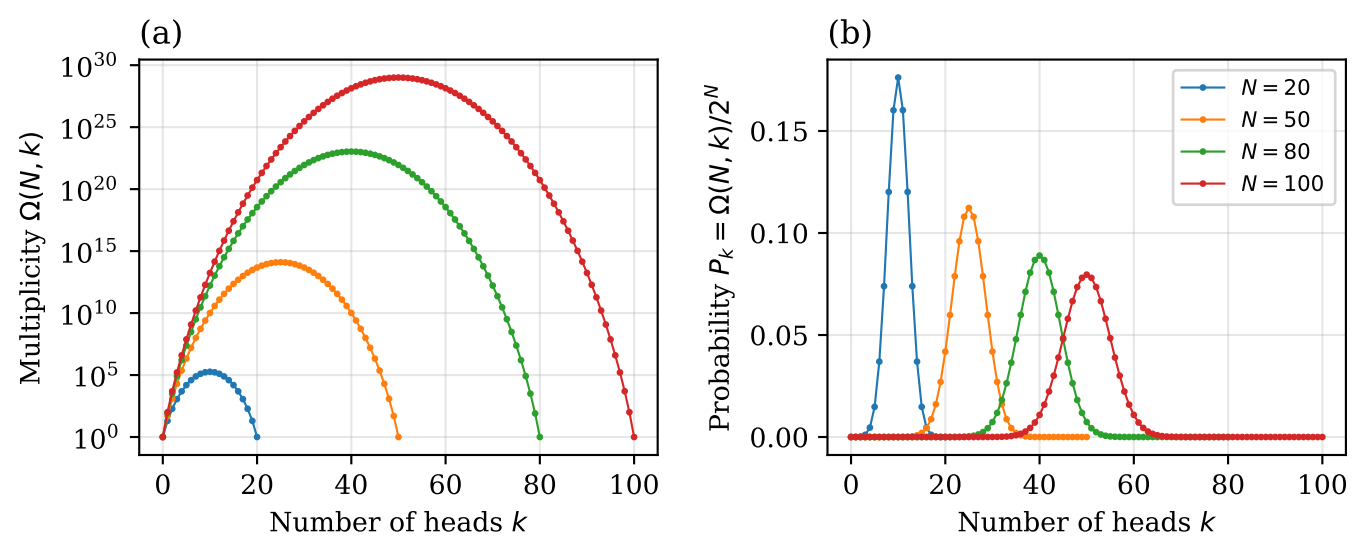
(a)
Multiplicity $\Omega(N, k)$
10^{30}
10^{25}
10^{20}
10^{15}
10^{10}
10^{5}
10^{0}
0
20
40
60
80
100
Number of heads k
(b)
Probability $P_k = \Omega(N, k)/2^N$
0.15
0.10
0.05
0.00
$N = 20$
$N = 50$
$N = 80$
$N = 100$
0
20
40
60
80
100
Number of heads k